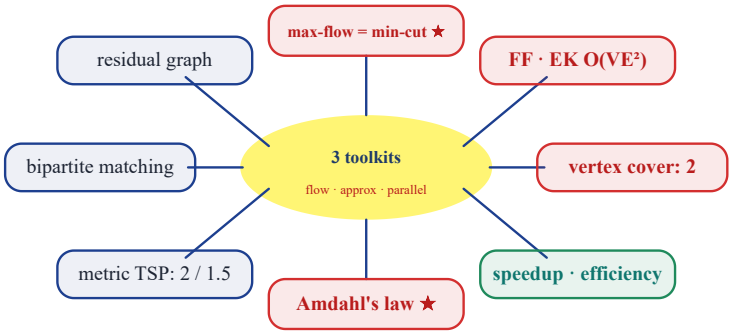
# Flow, Approximation & Parallel Algorithms

**max-flow = min-cut · Ford–Fulkerson · Edmonds–Karp · ρ-approximation · PTAS/FPTAS · speedup · Amdahl**

**Eleven sheets:** how much can a network carry and where it chokes, how to get a provably good answer when the exact one is out of reach, and how fast many processors really make a program — the three "beyond the basics" clauses of the algorithms syllabus.




↑ the whole topic: red = high-yield, green = the parallel metrics

## Syllabus coverage

The algorithms syllabus lists, after graph traversal, spanning trees and shortest paths: "**Flow algorithms; Approximation algorithms; Parallel algorithms;**" — followed by lower-bound theory and NP-completeness. **All three clauses are taken in full here.** Graph searching (BFS/DFS), MST and greedy methods come from the previous lesson; why some problems need approximation at all — **NP-hardness** — is the next lesson. Here the question is practical: **how much, how close, how fast?**

**High-yield list**

① $0 \le f \le c$; in = out except at s, t
② **max flow = min cut**
③ Residual: forward c−f, backward f
④ FF $O(E \cdot |f^*|)$, integer capacities
⑤ Edmonds–Karp = BFS paths, $O(VE^2)$
⑥ Dinic $O(V^2E)$ · push–relabel $O(V^2E)$
⑦ Matching = flow with unit capacities
⑧ $\rho \ge 1$; vertex cover 2-approx
⑨ Metric TSP 2 (MST), Christofides 1.5
⑩ Set cover greedy $H(n) \approx \ln n$
⑪ $S = T_1/T_p$, $E = S/p$
⑫ **Amdahl**: $S \le 1/((1-P)+P/N)$

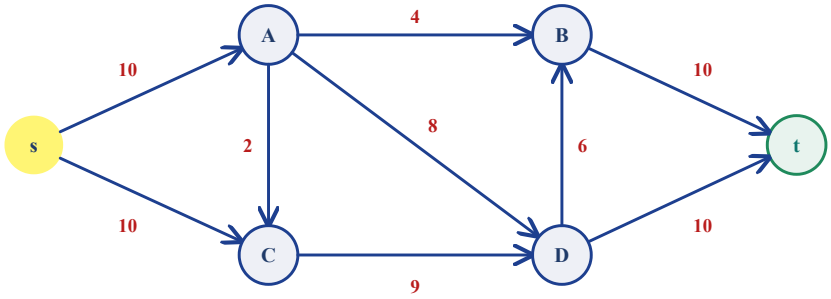
**Legend:** **yellow** = key term · **red circle** = must-memorise · **purple** = memory trick · **red** = exam trap · **green** = worked example

## Contents


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# 1 Flow Networks: Source, Sink, Capacity, Flow — and the Residual Graph

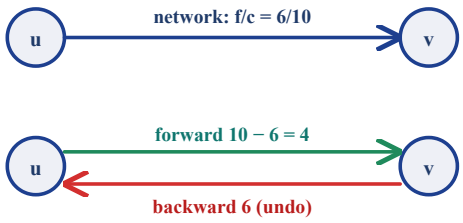
**Definition:** A **flow network** is a **directed graph G = (V, E) in which every edge (u, v) has a capacity c(u, v) ≥ 0**, with two special vertices: the **source s**, where flow originates, and the **sink t**, where it is absorbed. A **flow** f assigns to each edge the amount actually passing through it. Water pipes, road lanes, network links and conveyor belts are all the same model: *how much can travel from s to t per unit time?*




the running example of this lesson: red numbers are capacities; it is solved on sheet 4

| Rule | Statement | Meaning |
|---|---|---|
| **Capacity constraint** | **0 ≤ f(u,v) ≤ c(u,v)** | No edge carries more than its capacity, and never a negative amount. |
| **Flow conservation** | Σ f(·, v) = Σ f(v, ·) for every v ≠ s, t | **What enters a vertex leaves it** — only s and t are exempt. |
| **Value of the flow** | \|f\| = Σ f(s, v) − Σ f(v, s) | Net flow out of the source = net flow into the sink. |
| **Bottleneck** | c_f(p) = min residual capacity on path p | The amount one augmenting path can add. |

**Residual graph G_f ✍️:** For every edge carrying f of capacity c, the residual graph has a **forward edge** with **c_f(u,v) = c(u,v) − f(u,v)** (room left) and a **backward edge** with **c_f(v,u) = f(u,v)** (flow that may be cancelled). An **augmenting path** is any s⟶t path in G_f; edges with c_f = 0 are dropped.




one edge ⟶ two residual edges

**Exam trap ✍️ three:** **1.** Conservation does **not** hold at s and t — that is where the flow value is measured. **2.** The backward edge is **not a real pipe**; pushing along it means *cancelling* flow already sent. **3.** A saturated edge (f = c) disappears forward but **still exists backward**.

**Memory trick:** "**Forward = room left, backward = regret.**" Every unit you send buys you the right to change your mind about it later — that is the residual graph in one line.

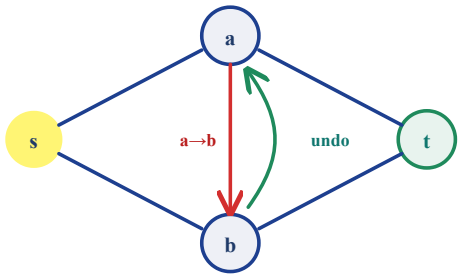
# 2 Ford–Fulkerson, Edmonds–Karp, Dinic and Push–Relabel

```
FORD-FULKERSON(G, s, t)
  for each edge (u,v): f(u,v) = 0
  while an s→t path p exists in G_f:
    b = min{ c_f(u,v) : (u,v) on p }
    for each (u,v) on p:
      if (u,v) ∈ E: f(u,v) += b
      else:         f(v,u) -= b
  return f      // |f| is maximum
```

**The method in four steps:** **1.** Start with **zero flow**. **2.** Find any augmenting path in the residual graph. **3.** Push its **bottleneck** b along it and update residual capacities (forward −b, backward +b). **4.** Repeat until **no s→t path remains**; the flow is then maximum. FF is a *method*: it does not say **how** to find the path.

| Algorithm | How the path / flow is chosen | Time |
|---|---|---|
| **Ford–Fulkerson** (1956) | Any augmenting path (often DFS). Needs **integer** capacities to be sure of stopping. | **O(E·\|f*\|)** |
| **Edmonds–Karp** (1972) | FF with **BFS = shortest augmenting path** (fewest edges). O(VE) augmentations × O(E) per BFS. | **O(VE²)** |
| **Dinic** (1970) | BFS builds a **level graph**; DFS pushes a **blocking flow**; at most V−1 phases. | O(V²E); unit capacity O(E·min(V^⅔, E^½)) |
| **Push–Relabel** (Goldberg–Tarjan) | Not path-based: keeps a **preflow**, pushes excess downhill, **relabels** heights. | generic O(V²E); FIFO O(V³) |
| **Hopcroft–Karp** | Dinic specialised to **bipartite matching**. | O(E√V) |

**Exam trap ✍ three corrections:** **1. Dinic is O(V²E)** in general. The bound O(E·min(V^⅔, E^½)) holds only for unit-capacity networks — quoting it as the general bound is wrong. **2. O(E·|f*|)** assumes integer capacities: each augmentation adds ≥ 1. With irrational capacities plain FF may **never terminate** (and may even converge to a wrong value). **3.** Edmonds–Karp's bound is **independent of the capacities** — that is exactly why BFS is used.



all capacities 1 · max flow 2

s→a→b→t first: only the backward edge b→a lets s→b→a→t reach flow 2

**Memory trick:** "**Ford finds any road, Edmonds–Karp takes the shortest, Dinic floods by levels, push–relabel lets water run downhill.**" Complexities pair up: **E-K = VE²**, **Dinic = V²E** — the squares swap places.

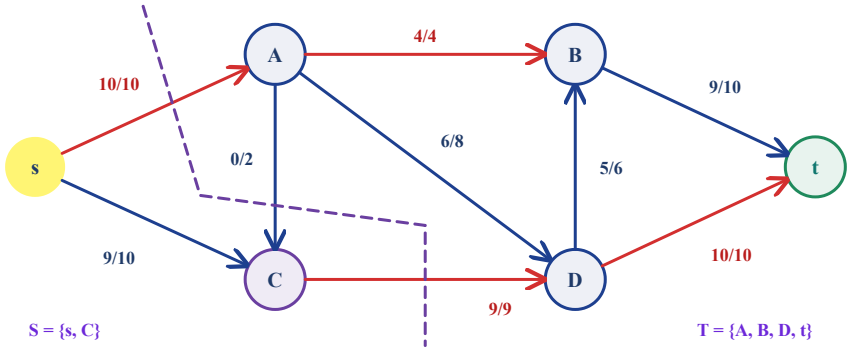
**Integrality theorem:** With **integer** capacities, FF returns a max flow in which **every edge carries an integer**. This is what makes the matching reduction (sheet 5) give a real matching, not fractions.

**Worked example ✍ why the path choice matters:** s→u 1000, s→v 1000, u→v 1, u→t 1000, v→t 1000. A careless FF can alternate s→u→v→t and s→v→u→t (through the backward edge v→u), adding **1** each time: **2000 augmentations**. Edmonds–Karp picks the 2-edge paths s→u→t and s→v→t first: **2 augmentations**, same max flow **2000**.

## 3 Worked Example: Max Flow of the Sheet-2 Network, Checked by a Min Cut

**Worked example ✍ Edmonds–Karp (shortest paths first):** Network: s→A 10, s→C 10, A→B 4, A→C 2, A→D 8, C→D 9, D→B 6, B→t 10, D→t 10. Each row: the path found in the current residual graph, its bottleneck, and what it saturates.

| # | Augmenting path | Residual capacities on the path | Bottleneck | Saturates | \|f\| |
|---|---|---|---|---|---|
| 1 | s→A→B→t | 10, 4, 10 | **4** | A→B | 4 |
| 2 | s→A→D→t | 6, 8, 10 | **6** | s→A | 10 |
| 3 | s→C→D→t | 10, 9, 4 | **4** | D→t | 14 |
| 4 | s→C→D→B→t | 6, 5, 6, 6 | **5** | C→D | **19** |
| 5 | From s only C is reachable (s→A = 0, C→D = 0, no flow on A→C to cancel) — **no augmenting path: stop.** | | | | |




final flow f/c (red = saturated); the purple dashed line is the minimum cut

| Cut S \| T | Forward edges S→T | c(S,T) |
|---|---|---|
| {s} | 10 + 10 | 20 |
| {s, A, C} | 4 + 8 + 9 | 21 |
| {s,A,B,C,D} | 10 + 10 | 20 |
| **{s, C}** | s→A 10 + C→D 9 | **19** |

**Check every vertex:** A: in 10, out 4+0+6 = 10 ✓ · C: in 9, out 9 ✓ · D: in 6+9 = 15, out 5+10 = 15 ✓ · B: in 4+5 = 9, out 9 ✓ · out of s 10+9 = **19** = into t 9+10 ✓. **The vertices reachable from s in the final residual graph form the min-cut side S.**

**Exam trap ✍ two:** **1.** A→C crosses the cut from T to S, so it is not added to the cut capacity — a cut counts only **S→T** edges. **2.** The obvious cuts ({s} = 20, into t = 20) are **not** the minimum; always search the residual graph for the true S.

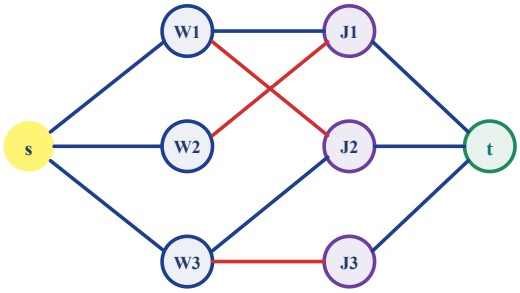
**Reading the answer like an engineer:** The min cut names the **bottleneck links**: only raising **s→A** or **C→D** can increase the throughput. Upgrading B→t or D→t to 100 changes nothing, because **the flow is limited by the cut, not by the sink side**. Edges outside every min cut have slack.

# 4 The Max-Flow Min-Cut Theorem, and Where Flow Is Used

**Definitions:** An **s–t cut** (S, T) splits V into two parts with s ∈ S and t ∈ T. Its **capacity** is **c(S,T) = Σ c(u,v) over edges u ∈ S, v ∈ T** — forward edges only. The **net flow across** it is forward flow minus backward flow, and for *every* cut it equals |f|.

**Max-flow min-cut theorem (Ford–Fulkerson, 1956) ✍️:** The following are **equivalent**: **(1)** f is a maximum flow; **(2)** the residual graph $G_f$ has **no augmenting path**; **(3)** |f| = c(S, T) for some cut. Hence **max |f| = min c(S,T)**. The easy half (**weak duality**): *every* flow ≤ *every* cut, because all flow must cross the cut.

**Bipartite matching as a flow ✍️:** Left set L, right set R. Add **s → every u ∈ L** and **every v ∈ R → t**, orient each L–R edge left to right, and give **every edge capacity 1**. By the integrality theorem the max flow is 0/1 on each edge, and the edges L→R carrying flow form a **maximum matching**: **max matching = max flow**.



red = matched (cap 1 each)

matching = unit-capacity flow

**Worked example:** Workers W1, W2, W3; jobs J1, J2, J3. W1 can do J1, J2; W2 only J1; W3 J2, J3. Paths s→W1→J1→t, s→W3→J2→t give 2; then s→W2→J1 is blocked forward, but the residual edge **J1→W1** lets s→W2→J1→W1→J2→W3→J3→t re-route: flow **3** = W1–J2, W2–J1, W3–J3.

| Application | What s, t, edges and capacities stand for |
|---|---|
| **Network routing / bandwidth** | Routers are vertices, links carry Mbps capacities; max flow = the most traffic s can send to t; the min cut is the **set of links whose failure disconnects them cheapest**. |
| **Transportation & logistics** | Depots, roads with truck limits, warehouses — how many tonnes per day can reach the port. |
| **Image segmentation** | Pixels are vertices; s = "foreground", t = "background"; edge weights = similarity. The **min cut** is the cheapest boundary between object and background (graph cuts). |
| **Disjoint paths (Menger)** | Unit capacities: max flow = number of **edge-disjoint** s–t paths = fewest links whose removal disconnects s from t. |
| **Scheduling / assignment** | Crews to flights, students to projects — matching with capacities; also project selection and baseball elimination. |

**Exam trap:** Max flow equals the **minimum** cut, not "the sum of all capacities" nor "the largest edge". A flow network can have several minimum cuts but only one max-flow *value*.

**Memory trick:** **"A chain is as strong as its weakest link — a network is as wide as its narrowest cut."**

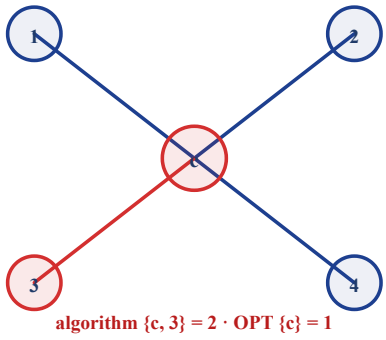
# 5 Approximation Algorithms: the Ratio ρ, and Vertex Cover

**Definition:** An **approximation algorithm** runs in **polynomial time** on an **NP-hard optimisation problem** and returns a solution with a **proven guarantee** of how far it can be from the optimum. (Why the exact answer is out of reach — NP-hardness — is the subject of the next lesson.) A heuristic with no guarantee is *not* an approximation algorithm.

| Problem type | Approximation ratio ρ(n) | Reading it |
|---|---|---|
| **Minimisation** (cover, tour) | **$\rho = C / C^* \geq 1$** | C = algorithm's cost, C* = optimal. ρ = 2: never more than twice the optimum. |
| **Maximisation** (knapsack, cut) | $\rho = C^* / C \geq 1$ | ρ = 2: never less than half the optimum. |
| **Combined (CLRS)** | $\max(C/C^*, C^*/C) \leq \rho(n)$ | ρ = 1 means exact; the nearer to 1, the better. |

| Problem | Algorithm | Ratio | Note |
|---|---|---|---|
| **Vertex cover** | take both ends of a maximal matching | **2** | sheet below |
| **Metric TSP** | MST, preorder walk, shortcut | **2** | Christofides **1.5** |
| **Set cover** | greedy: most uncovered elements | **$H(n) \approx \ln n$** | $\leq \ln n + 1$ |
| **0/1 knapsack** | value-scaling DP | **$1 + \varepsilon$** | FPTAS |

```
APPROX-VERTEX-COVER(G)
  C = ∅;  E' = E
  while E' ≠ ∅:
    pick any edge (u,v) in E'
    C = C ∪ {u, v}
    delete from E' every edge touching u or v
  return C            // O(V + E)
```




a star: the bound 2 is tight

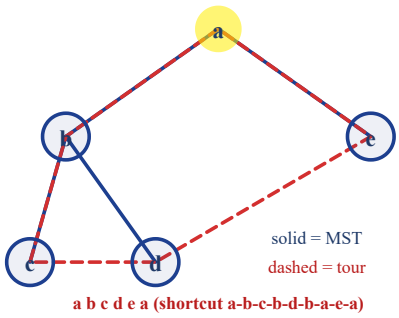
**Why ratio 2 ✍ proof:** The picked edges form a **matching M** (no two share a vertex), and |C| = 2|M|. Any cover — the optimal C* too — must contain **at least one endpoint of each edge of M**, and those endpoints are all different, so |C*| ≥ |M|. Hence **$|C| = 2|M| \leq 2|C^*|$**.

**Exam trap ✍ two:** **1.** The "pick the highest-degree vertex" greedy is **not** a 2-approximation — its ratio grows like log n. The 2 belongs to the **both-endpoints** rule. **2.** A *minimisation* ratio of 2 allows up to 2 × OPT, never below OPT.

**Memory trick:** "**Grab the whole edge, pay double at most.**" One endpoint of each grabbed edge was needed anyway; the other is the price of not knowing which.

## 6 Metric TSP, Set Cover, Knapsack — and PTAS vs FPTAS

**Metric TSP: the MST tour ✍:** Distances obey the **triangle inequality** d(u,w) ≤ d(u,v) + d(v,w). **1.** Build an **MST** T (Prim/Kruskal, previous lesson). **2.** Walk it in **DFS preorder**: the full walk uses each MST edge twice, cost 2·w(T). **3. Shortcut** repeated vertices — by the triangle inequality this never adds cost. Since deleting one edge of the optimal tour leaves a spanning tree, **w(T) ≤ OPT**, so tour ≤ 2·w(T) ≤ **2·OPT**.




preorder a, b, c, d, e, then back to a

**Beyond 2 — Christofides (1976):** MST + a **minimum-weight perfect matching** on the odd-degree MST vertices gives an Eulerian graph; shortcut its Euler tour → **ratio 1.5** for metric TSP. **General TSP** (no triangle inequality) has no constant-ratio approximation unless P = NP.

**Set cover: greedy:** Universe U of n elements, a family of subsets; choose the fewest subsets covering U. Greedy: **repeatedly take the set covering the most still-uncovered elements**. Ratio **H(n) = 1 + ½ + … + 1/n ≤ ln n + 1** — essentially the best possible in polynomial time (unless P = NP).

**Worked example:** U = {1,…,6}; S1 = {1,2,3,4}, S2 = {4,5}, S3 = {5,6}, S4 = {1,6}. Greedy takes S1 (4 new), then S3 (5, 6: 2 new) → **2 sets**, all covered. Here greedy is optimal; on adversarial inputs it can be ≈ ln n times worse.

**0/1 knapsack: FPTAS ✍:** The exact DP is O(nW) — **pseudo-polynomial**. Scale every value down: v′ = ⌊v / K⌋ with **K = ε·v_max / n**, run the DP indexed by value in O(n³/ε), and the answer is ≥ **(1 − ε)·OPT**. Simpler: max(greedy by value/weight, best single item) is a **2-approximation**. (Fractional knapsack needs none of this — greedy is exact.)

| Scheme | Guarantee | Running time | Example |
|---|---|---|---|
| **PTAS** | within (1 + ε) for **every fixed** ε > 0 | polynomial in n, may be exponential in 1/ε, e.g. O(n^{1/ε}) | Euclidean TSP, bin-packing variants |
| **FPTAS** | within (1 + ε) | **polynomial in n and 1/ε**, e.g. O(n³/ε) | 0/1 knapsack, subset sum |

**Exam trap ✍ three:** **1.** Every FPTAS is a PTAS, **not** the reverse. **2.** A strongly NP-hard problem has no FPTAS unless P = NP; knapsack is only *weakly* NP-hard. **3.** "Set cover greedy ratio 2" is wrong — it is **ln n**.

**Memory trick:** "**Cover 2, Tour 2 then 1.5, Set ln n, Knapsack 1+ε.**" And "**F**PTAS is **F**ully polynomial — in **both** n and 1/ε."

## 7 Parallel Algorithms: Speedup, Efficiency, Granularity, Amdahl's Law

**Definition:** A **parallel algorithm divides a problem into sub-problems solved simultaneously on several processors**, then combines the results. Its quality is measured against the **best sequential** algorithm, not against itself on one processor.

| Measure | Formula | Meaning |
|---|---|---|
| **Speedup S** | **$S = T_1 / T_p$** | $T_1$ = best sequential time, $T_p$ = time on p processors. Ideal (linear): S = p. |
| **Efficiency E** | $E = S / p$ | Fraction of processor time doing useful work; $0 < E \leq 1$ normally. |
| **Cost** | $p \times T_p$ | **Cost-optimal** if $p \cdot T_p = \Theta(T_1)$. |
| **Granularity** | computation / communication | **Fine**-grained: many small tasks, much communication. **Coarse**-grained: few large tasks. |
| **Work–span** | $T_p \geq T_1/p$, $T_p \geq T\infty$ | $T\infty$ (span) = longest dependency chain; parallelism = $T_1/T\infty$; Brent: $T_p \leq T_1/p + T\infty$. |

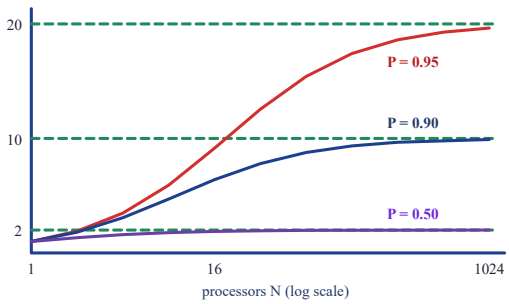
**Data vs task parallelism:** **Data**: the **same operation on different pieces of data** — vector addition, matrix multiply, image filters (SIMD, GPUs). **Task**: **different operations at once** — pipeline stages, a web server's independent requests (MIMD).

**Flynn's taxonomy:** **SISD** classic uniprocessor · **SIMD** vector units, GPUs · **MISD** rare (fault-tolerant pipelines) · **MIMD** multicores, clusters. PRAM models: **EREW**, **CREW**, **CRCW** (exclusive/concurrent read/write).

**Amdahl's law (fixed problem size) ✍️:** If a fraction **P** of the work can run in parallel on **N** processors and 1 − P stays serial: **$S(N) = 1 / ((1 - P) + P/N)$**. As $N \to \infty$, **$S \to 1 / (1 - P)$** — the serial part caps the speedup.

**Worked example:** **P = 0.9, N = 10:** S = 1 / (0.1 + 0.09) = 1/0.19 ≈ **5.26**; E = 5.26/10 ≈ **53%**; limit 1/0.1 = **10**, however many processors. **P = 0.8, N = 4:** S = 1/(0.2 + 0.2) = **2.5**; limit 5.

**Gustafson's law (scaled size):** When the problem grows with N: **$S = (1 - P) + P \cdot N$** (P = parallel share of the parallel run). P = 0.9, N = 10 → 0.1 + 9 = **9.1**. Amdahl is pessimistic, Gustafson optimistic.




speedup flattens at 1/(1−P): 2, 10, 20

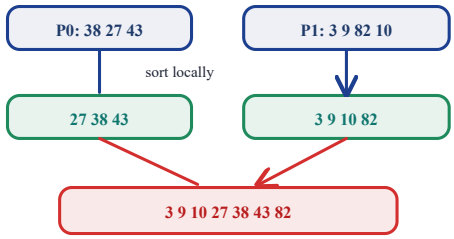
**Exam trap:** In Amdahl's law **P is the parallel fraction**; mixing it with the serial fraction inverts every answer. Speedup **above p** (superlinear) can occur through cache effects, but is not the norm.

**Memory trick:** **"The slowest part sets the pace."** Ten percent serial ⇒ never more than **ten times faster**.

## 8 Parallel Sorting, Matrix Multiplication, BFS — and the Pitfalls

| Problem | Parallel approach | Cost / time |
|---|---|---|
| **Reduction (sum of n)** | Pairwise tree: n/2 adds, then n/4, … | **O(log n)** steps, n/2 processors |
| **Parallel merge sort** | Split among p processors, sort locally, merge in a tree of log p levels. | O((n/p) log(n/p)) local sort + merging |
| **Bitonic sort** | Sorting **network**: build bitonic sequences, then bitonic-merge with compare–exchange; n a power of 2. | **O(log² n)** time, O(n log² n) comparisons |
| **Matrix multiply** | **Block** decomposition; **Cannon's** algorithm on a √p × √p processor grid. | O(n³/p) compute, √p shift steps |
| **Graph search** | **Level-synchronous BFS**: expand the whole frontier in parallel; also parallel Dijkstra (Δ-stepping). | rounds = depth of the BFS tree |

**Worked example: parallel merge sort:** Array [38, 27, 43, 3, 9, 82, 10] on 2 processors. **P0** gets [38, 27, 43] → sorts to [27, 38, 43]; **P1** gets [3, 9, 82, 10] → [3, 9, 10, 82] — **both at the same time**. One merge step → **[3, 9, 10, 27, 38, 43, 82]**. With p processors the merge tree has $\log_2 p$ levels.




split, sort in parallel, merge

**Bitonic half-cleaner ✍️:** A **bitonic** sequence rises then falls (or a rotation of one). Compare item i with item i + n/2 and swap if out of order: [2, 5, 8, 9 | 7, 6, 3, 1] → low [2, 5, 3, 1], high [7, 6, 8, 9]. Every low ≤ every high, and **each half is bitonic again** — recurse. n = 8 needs 3·4/2 = **6** compare stages.

**Cannon's algorithm:** Processor P(i,j) holds blocks A(i,j), B(i,j). **Skew**: shift row i of A left by i, column j of B up by j. Then √p times: **multiply-add** local blocks, shift A one left and B one up. Each processor stores **only one block of each matrix at a time**.

| Challenge | What goes wrong / the remedy |
|---|---|
| **Communication overhead** | Sending data costs time; fine granularity can make $T_p$ *worse* than $T_1$. |
| **Load balancing** | Uneven pieces leave processors idle — dynamic scheduling, work stealing. |
| **Synchronisation, races, deadlocks** | Barriers make fast processors wait; unordered updates race (BFS "visited" flag → atomic compare-and-swap); circular lock waits deadlock. |

**Applications:** Scientific simulation (weather, physics) · **big data**: Hadoop MapReduce, Apache Spark · graphics **rendering** on GPUs · **distributed ML** training (data-parallel gradients, all-reduce).

**Exam trap:** Bitonic sort does **more** comparisons than merge sort (n log² n vs n log n) — it wins because its compare pattern is fixed, ideal for hardware and GPUs.

# Night-Before-the-Exam Sheet

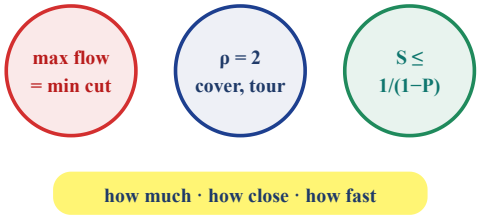
**Fact sheet:**

| | |
|---|---|
| $0 \le f(u,v) \le c(u,v)$ | In = out, except s and t |
| $\|f\|$ = net flow out of s | Residual fwd c − f, back f |
| Augmenting path: s→t in $G_f$ | Bottleneck = min $c_f$ on path |
| FF: $O(E \cdot \|f^*\|)$, integer caps | Irrational caps: may not stop |
| Edmonds–Karp: BFS, $O(VE^2)$ | Dinic: $O(V^2E)$ |
| Dinic unit caps: $E \cdot \min(V^{2/3}, E^{1/2})$ | Push–relabel $O(V^2E)$, FIFO $O(V^3)$ |
| **max flow = min cut** | Cut counts S→T edges only |
| Matching = unit-cap flow | Hopcroft–Karp $O(E\sqrt{V})$ |
| Min: $\rho = C/C^* \ge 1$ | Max: $\rho = C^*/C \ge 1$ |
| Vertex cover: 2 | Metric TSP: 2 (MST) |
| Christofides: 1.5 | General TSP: no constant |
| Set cover: $H(n) \le \ln n + 1$ | Knapsack: FPTAS $O(n^3/\varepsilon)$ |
| PTAS: poly in n | FPTAS: poly in n and $1/\varepsilon$ |
| $S = T_1/T_p \cdot E = S/p$ | Cost = $p \cdot T_p$ |
| Amdahl $1/((1-P)+P/N)$ | Limit $1/(1-P)$ |
| Gustafson $(1-P)+P \cdot N$ | Bitonic $O(\log^2 n)$ |

## Top 12 one-liners for MCQs

1. Max flow = capacity of a **min** cut
2. Edmonds–Karp = FF + **BFS**
3. E-K **$VE^2$**, Dinic **$V^2E$**
4. Back edges let flow be **cancelled**
5. Conservation fails only at s, t
6. Matching: all capacities **1**
7. Vertex cover: both endpoints, **2**
8. MST tour **2**, Christofides **1.5**
9. Greedy set cover **ln n**
10. FPTAS ⊂ PTAS
11. **P = 0.9 ⇒ S ≤ 10**
12. E = S/p, ideal E = 1

**Top traps:** quoting Dinic's **unit-capacity** bound as general · FF **always terminates** (only with integer/rational caps) · adding **T→S** edges to a cut · max-degree greedy as **2-approx** · set cover ratio **2** · **FPTAS** exponential in $1/\varepsilon$ · Amdahl with the **serial** fraction as P.




three numbers, three toolkits

**All tricks together:** Forward = room left, backward = regret · Ford any road, E-K shortest, Dinic levels, push–relabel downhill · As wide as the narrowest cut · Grab the whole edge, pay double · Cover 2, Tour 2 → 1.5, Set ln n, Knapsack 1+ε · The slowest part sets the pace.

## ✓ Practice MCQs (exam pattern, 10 marks)

**1.** By the max-flow min-cut theorem, the maximum s–t flow equals:
(a) The number of edges (b) The sum of all capacities
(c) The capacity of a minimum s–t cut (d) The largest edge capacity

**2.** Edmonds–Karp (FF with BFS augmenting paths) runs in:
(a) $O(VE^2)$ (b) $O(E \log V)$
(c) $O(V^2)$ (d) $O(E \cdot |f^*|)$ only

**3.** An edge with capacity 10 carries flow 6. Its residual edges are:
(a) forward 6, backward 4 (b) forward 10, backward 0
(c) forward 16, backward 6 (d) forward 4, backward 6

**4.** Taking both endpoints of a maximal matching gives a vertex cover within a factor of:
(a) 1.5 (b) 2
(c) log n (d) n

**5.** The MST-based (double-tree) algorithm for **metric** TSP has ratio:
(a) 2 (b) 1.5
(c) ln n (d) no constant

**6.** An FPTAS runs in time polynomial in:
(a) n only (b) 1/ε only
(c) both n and 1/ε (d) neither

**7.** By Amdahl's law, with 90% parallel code on 10 processors the speedup is about:
(a) 9.1 (b) 5.26
(c) 10 (d) 4.75

**8.** A program gets speedup 6 on 8 processors. Its efficiency is:
(a) 1.33 (b) 48
(c) 0.60 (d) 0.75

**9.** Bitonic sort with n/2 comparators per stage takes parallel time:
(a) $O(\log n)$ (b) $O(n)$
(c) $O(\log^2 n)$ (d) $O(n \log n)$

**10.** The greedy set-cover algorithm has approximation ratio:
(a) $H(n) \approx \ln n$ (b) 2
(c) $1 + \varepsilon$ (d) 1.5

Answer key → 1-c · 2-a · 3-d · 4-b · 5-a · 6-c · 7-b · 8-d · 9-c · 10-a

**Written question:** Define a **flow network** and state the capacity and conservation constraints. Find the maximum flow of the network on sheet 2 by Ford–Fulkerson with BFS paths, and verify it with a **minimum cut**. State **Amdahl's law** and compute the speedup for P = 0.8 on 4 processors and its upper limit. 10 marks

1. **Definition (2)** — directed graph, capacity $c(u,v) \geq 0$ on each edge, source s, sink t. **Capacity:** $0 \leq f(u,v) \leq c(u,v)$. **Conservation:** inflow = outflow at every vertex except s and t. 2
2. **Augmentations (3)** — s→A→B→t (4), s→A→D→t (6), s→C→D→t (4), s→C→D→B→t (5): total **19**; then no s→t path exists in the residual graph. 3
3. **Min cut (2)** — S = {s, C} (reachable from s in $G_f$), T = rest; S→T edges s→A (10) + C→D (9) = **19** = |f|, so the flow is maximum by the max-flow min-cut theorem. 2
4. **Amdahl (3)** — $S = 1/((1 - P) + P/N) = 1/(0.2 + 0.8/4) = 1/0.4 =$ **2.5**; efficiency 2.5/4 = 62.5%; limit $N \to \infty$: $1/(1 - 0.8) =$ **5** — the 20% serial part caps it. 3

## Before the exam ✓

- ✓ Can run Ford–Fulkerson by hand with residuals
- ✓ Can find the min cut from the final residual graph
- ✓ Can quote the four max-flow complexities
- ✓ Can prove the vertex-cover ratio 2
- ✓ Can separate PTAS from FPTAS
- ✓ Can compute Amdahl speedup, limit and efficiency